

Australian statement of hazardous nature : Classified as hazardous according to criteria of Safe Work Australia

## Section 1 - Identification

**Product Name** Lithium bis(trimethylsilyl)amide

**CAS No** 4039-32-1

**Synonyms** Lithium hexamethyldisilazide; 1,1,1,3,3,3-Hexamethyldisilazane lithium salt

**Product Code** **338140000; 338140100; 338140500; 338142500**

**Address** ThermoFisher Scientific Australia Pty Ltd  
 5 Caribbean Drive, Scoresby  
 VICTORIA 3179, Australia

**Emergency Tel.** **CHEMTREC®**  
**03 9757 4559 or +613 9757 4559**

**Telephone / Fax Numbers**  
 Tel: 1300 735 292  
 Fax: 1800 067 639

**E-mail address** ANZinfo@thermofisher.com

**Recommended Use** Laboratory chemicals.

**Uses advised against** This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list. This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction. This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern.

## Section 2 - Hazard(s) Identification

### Classification under Safe Work Australia

Classified as hazardous according to criteria of Safe Work Australia

#### Physical hazards

Flammable solids

Category 2

#### Health hazards

Skin Corrosion/Irritation  
 Serious Eye Damage/Eye Irritation

Category 1 B  
 Category 1

#### Environmental hazards

No hazards identified

### Label Elements



Flame



Corrosion

**Signal Word****Danger****Hazard Statements**

H228 - Flammable solid

H314 - Causes severe skin burns and eye damage

AUH014 - Reacts violently with water

**Precautionary Statements**

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

P240 - Ground and bond container and receiving equipment

P264 - Wash face, hands and any exposed skin thoroughly after handling

P280 - Wear protective gloves/protective clothing/eye protection/face protection

P301 + P330 + P331 - IF SWALLOWED: rinse mouth. Do NOT induce vomiting

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P310 - Immediately call a POISON CENTER or doctor

P363 - Wash contaminated clothing before reuse

P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

P403 - Store in a well-ventilated place

P501 - Dispose of contents/ container to an approved waste disposal plant

**Other information**

## Section 3 - Composition and Information on Ingredients

| Component                                                     | CAS No    | Weight % |
|---------------------------------------------------------------|-----------|----------|
| Silanamine, 1,1,1-trimethyl-N-(trimethylsilyl)-, lithium salt | 4039-32-1 | > 94     |

## Section 4 - First Aid Measures

**Inhalation**

Remove to fresh air. If breathing is difficult, give oxygen. Call a physician or poison control center immediately. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device.

**Ingestion**

Do NOT induce vomiting. Never give anything by mouth to an unconscious person. Drink plenty of water. Immediate medical attention is required.

**Skin Contact**

Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Call a physician immediately.

**Eye Contact**

Immediate medical attention is required. Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Keep eye wide open while rinsing.

|                                            |                                                                                                                                                                                                                                                                                                   |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                      | Immediate medical attention is required. Show this safety data sheet to the doctor in attendance.                                                                                                                                                                                                 |
| <b>Self-Protection of the First Aider</b>  | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.                                                                                                                                                  |
| <b>First Aid Facilities</b>                | Eyewash, safety shower and washroom.                                                                                                                                                                                                                                                              |
| <b>Most important symptoms and effects</b> | Causes burns by all exposure routes. . Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation |
| <b>Notes to Physician</b>                  | Treat symptomatically.                                                                                                                                                                                                                                                                            |

## Section 5 - Fire Fighting Measures

### Suitable Extinguishing Media

CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam.

### Extinguishing media which must not be used for safety reasons

Water.

### Hazardous Decomposition Products

Nitrogen oxides (NO<sub>x</sub>), Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Silicon dioxide, Ammonia, Thermal decomposition can lead to release of irritating gases and vapors.

### Specific Hazards Arising from the Chemical

The product causes burns of eyes, skin and mucous membranes. Reacts violently with water. Flammable.

### Special protective equipment and precautions for fire fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## Section 6 - Accidental Release Measures

### Emergency procedures

Evacuate personnel to safe areas. Use personal protective equipment as required. Avoid contact with skin, eyes or clothing.

### Environmental Precautions

Do not allow material to contaminate ground water system. Should not be released into the environment. See Section 12 for additional Ecological Information.

### Methods for Containment and Clean Up

#### Clean-up methods - small spillage

Sweep up and shovel into suitable containers for disposal. Avoid dust formation. Do not expose spill to water.

#### Clean-up methods - large spillage

Typically only supplied in small quantities as packaged goods.

If extremely toxic or used in larger quantities ensure a spillage action plan is in place. Evacuate area. Control the source and/or contain the spill if safe and able to do so. Use temporary diking, sand bags, dry sand, earth or proprietary booms/absorbent pads if available. Obtain advice on containment, neutralisation and clean-up from local emergency responders.

### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## Section 7 - Handling and Storage

**Precautions for Safe Handling**

Do not breathe dust. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Wear personal protective equipment/face protection. Do not ingest. If swallowed then seek immediate medical assistance. Do not allow contact with water.

**Conditions for Safe Storage, Including any Incompatibilities**

Keep containers tightly closed in a dry, cool and well-ventilated place. Corrosives area. Keep away from water or moist air. Store under an inert atmosphere. Flammables area.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

## Section 8 - Exposure Controls and Personal Protection

**Exposure limits**

DE - MAK and BAT values of Hazardous Chemical Compounds in the Work Area. Published by German Research Foundation on July 1, 2011

| Component                                                               | Australia | New Zealand WEL | ACGIH TLV | The United Kingdom | Germany                                                                                                                                                                                                                                                                                                                                         |
|-------------------------------------------------------------------------|-----------|-----------------|-----------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Silanamine,<br>1,1,1-trimethyl-N-(tri<br>methylsilyl)-, lithium<br>salt |           |                 |           |                    | TWA: 0.2 mg/m <sup>3</sup> (8<br>Stunden). MAK<br>inorganic compounds,<br>except Lithium and<br>strong irritant Lithium<br>compounds such as<br>Lithium amide, Lithium<br>hydride, Lithium<br>hydroxide, Lithium<br>nitride, Lithium oxide,<br>Lithium tetrahydro<br>aluminate, Lithium<br>tetrahydroborate<br>Höhepunkt: 0.2 mg/m <sup>3</sup> |

**Biological limit values**

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

**Exposure Controls****Engineering Measures**

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Personal protective equipment****Eye Protection**

Goggles (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

**Hand Protection**

Protective gloves

| Glove material | Breakthrough time | Glove thickness | AUS/NZ Standard | Glove comments        |
|----------------|-------------------|-----------------|-----------------|-----------------------|
| Nitrile rubber | See manufacturers | -               | AS/NZS 2161     | (minimum requirement) |
| Neoprene       | recommendations   |                 |                 |                       |
| Natural rubber |                   |                 |                 |                       |
| PVC            |                   |                 |                 |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger

of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Long sleeved clothing

**Respiratory Protection** Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices

**Recommended Filter type:** Particulates filter conforming to EN 143 (or AUS/NZ equivalent)

**Recommended half mask:-** Particle filtering: EN149:2001 (or AUS/NZ equivalent)

When RPE is used a face piece Fit Test should be conducted

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls** No information available.

## Section 9 - Physical and Chemical Properties

### Information on basic physical and chemical properties

|                                                |                                  |                                          |
|------------------------------------------------|----------------------------------|------------------------------------------|
| <b>Appearance</b>                              | Light yellow                     |                                          |
| <b>Physical State</b>                          | Solid                            |                                          |
| <b>Odor</b>                                    | Slight                           |                                          |
| <b>Odor Threshold</b>                          | No data available                |                                          |
| <b>pH</b>                                      | Not applicable                   |                                          |
| <b>Melting Point/Range</b>                     | No data available                |                                          |
| <b>Softening Point</b>                         | No data available                |                                          |
| <b>Boiling Point/Range</b>                     | 114 - 116 °C / 237.2 - 240.8 °F  |                                          |
| <b>Flash Point</b>                             | No information available         | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>                        | Not applicable                   | Solid                                    |
| <b>Flammability (solid,gas)</b>                | No information available         |                                          |
| <b>Explosion Limits</b>                        | No data available                |                                          |
| <b>Vapor Pressure</b>                          | No data available                |                                          |
| <b>Vapor Density</b>                           | Not applicable                   | Solid                                    |
| <b>Specific Gravity / Density</b>              | No data available                |                                          |
| <b>Bulk Density</b>                            | No data available                |                                          |
| <b>Water Solubility</b>                        | Decomposes in contact with water |                                          |
| <b>Solubility in other solvents</b>            | No information available         |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                                  |                                          |
| <b>Autoignition Temperature</b>                | No data available                |                                          |
| <b>Decomposition Temperature</b>               | No data available                |                                          |
| <b>Viscosity</b>                               | Not applicable                   | Solid                                    |
| <b>Explosive Properties</b>                    | No information available         |                                          |
| <b>Oxidizing Properties</b>                    | No information available         |                                          |
| <b>Other information</b>                       |                                  |                                          |
| <b>Molecular Formula</b>                       | C6 H18 Li N Si2                  |                                          |
| <b>Molecular Weight</b>                        | 167.33                           |                                          |

## Section 10 - Stability and Reactivity

**Reactivity** Yes

**Stability** Air sensitive.

**Conditions to Avoid** Incompatible products, Excess heat, Exposure to moisture, Exposure to air.

|                                         |                                                                                                                                                                                                  |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Incompatible Materials</b>           | Acids, Strong oxidizing agents, Alcohols, Halogens, Ketones, Carbon dioxide (CO <sub>2</sub> ).                                                                                                  |
| <b>Hazardous Decomposition Products</b> | Nitrogen oxides (NO <sub>x</sub> ). Carbon monoxide (CO). Carbon dioxide (CO <sub>2</sub> ). Silicon dioxide. Ammonia. Thermal decomposition can lead to release of irritating gases and vapors. |
| <b>Hazardous Polymerization</b>         | No information available.                                                                                                                                                                        |

## Section 11 - Toxicological Information

### Information on Toxicological Effects

|                                                   |                                                                                                                                                                                                                                                            |
|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Product Information</b>                        | No acute toxicity information is available for this product                                                                                                                                                                                                |
| <b>(a) acute toxicity;</b>                        |                                                                                                                                                                                                                                                            |
| <b>Oral</b>                                       | No data available                                                                                                                                                                                                                                          |
| <b>Dermal</b>                                     | No data available                                                                                                                                                                                                                                          |
| <b>Inhalation</b>                                 | No data available                                                                                                                                                                                                                                          |
| <b>(b) skin corrosion/irritation;</b>             | Category 1 B                                                                                                                                                                                                                                               |
| <b>(c) serious eye damage/irritation;</b>         | Category 1                                                                                                                                                                                                                                                 |
| <b>(d) respiratory or skin sensitization;</b>     |                                                                                                                                                                                                                                                            |
| <b>Respiratory</b>                                | No data available                                                                                                                                                                                                                                          |
| <b>Skin</b>                                       | No data available                                                                                                                                                                                                                                          |
| <b>(e) germ cell mutagenicity;</b>                | No data available                                                                                                                                                                                                                                          |
| <b>(f) carcinogenicity;</b>                       | No data available<br>There are no known carcinogenic chemicals in this product                                                                                                                                                                             |
| <b>(g) reproductive toxicity;</b>                 | No data available                                                                                                                                                                                                                                          |
| <b>(h) STOT-single exposure;</b>                  | No data available                                                                                                                                                                                                                                          |
| <b>(i) STOT-repeated exposure;</b>                | No data available                                                                                                                                                                                                                                          |
| <b>Target Organs</b>                              | No information available.                                                                                                                                                                                                                                  |
| <b>(j) aspiration hazard;</b>                     | Not applicable<br>Solid                                                                                                                                                                                                                                    |
| <b>Other Adverse Effects</b>                      | The toxicological properties have not been fully investigated.                                                                                                                                                                                             |
| <b>Symptoms / effects, both acute and delayed</b> | Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation |

## Section 12 - Ecological Information

|                                      |                                                                          |
|--------------------------------------|--------------------------------------------------------------------------|
| <b>Ecotoxicity effects</b>           | Reacts with water so no ecotoxicity data for the substance is available. |
| <b>Persistence and Degradability</b> | No information available                                                 |
| <b>Persistence</b>                   | Persistence is unlikely, based on information available.                 |

|                                              |                                                                           |
|----------------------------------------------|---------------------------------------------------------------------------|
| <b>Degradability</b>                         | Decomposes in contact with water.                                         |
| <b>Degradation in sewage treatment plant</b> | No information available. Decomposes in contact with water.               |
| <b>Bioaccumulative Potential</b>             | Product does not bioaccumulate due to reaction with water                 |
| <b>Mobility</b>                              | Decomposes in contact with water. Is not likely mobile in the environment |
| <b>Endocrine Disruptor Information</b>       | This product does not contain any known or suspected endocrine disruptors |
| <b>Persistent Organic Pollutant</b>          | This product does not contain any known or suspected substance            |
| <b>Ozone Depletion Potential</b>             | This product does not contain any known or suspected substance            |

## Section 13 - Disposal Considerations

|                                            |                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste from Residues/Unused Products</b> | Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.                                                                  |
| <b>Contaminated Packaging</b>              | Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.                                                                                                                                              |
| <b>Other Information</b>                   | Chemical wastes should be disposed through a licensed commercial waste collection service. Waste codes should be assigned by the user based on the application for which the product was used. Do not flush to sewer. Can be landfilled or incinerated, when in compliance with local regulations. Do not empty into drains. Large amounts will affect pH and harm aquatic organisms. |

## Section 14 - Transport Information

### IMDG/IMO

|                                |                                                               |
|--------------------------------|---------------------------------------------------------------|
| <b>UN-No</b>                   | UN2925                                                        |
| <b>Proper Shipping Name</b>    | Flammable solid, corrosive, organic, n.o.s.                   |
| <b>Technical Shipping Name</b> | Silanamine, 1,1,1-trimethyl-N-(trimethylsilyl)-, lithium salt |
| <b>Hazard Class</b>            | 4.1                                                           |
| <b>Subsidiary Hazard Class</b> | 8                                                             |
| <b>Packing Group</b>           | II                                                            |

### ADG

|                                |                                                               |
|--------------------------------|---------------------------------------------------------------|
| <b>UN-No</b>                   | UN2925                                                        |
| <b>Proper Shipping Name</b>    | Flammable solid, corrosive, organic, n.o.s.                   |
| <b>Technical Shipping Name</b> | Silanamine, 1,1,1-trimethyl-N-(trimethylsilyl)-, lithium salt |
| <b>Hazard Class</b>            | 4.1                                                           |
| <b>Subsidiary Hazard Class</b> | 8                                                             |
| <b>Packing Group</b>           | II                                                            |

### IATA

|                                |                                                               |
|--------------------------------|---------------------------------------------------------------|
| <b>UN-No</b>                   | UN2925                                                        |
| <b>Proper Shipping Name</b>    | Flammable solid, corrosive, organic, n.o.s.                   |
| <b>Technical Shipping Name</b> | Silanamine, 1,1,1-trimethyl-N-(trimethylsilyl)-, lithium salt |
| <b>Hazard Class</b>            | 4.1                                                           |
| <b>Subsidiary Hazard Class</b> | 8                                                             |
| <b>Packing Group</b>           | II                                                            |

|                              |                                 |
|------------------------------|---------------------------------|
| <b>Environmental hazards</b> | No hazards identified           |
| <b>Special Precautions</b>   | No special precautions required |

Additional information None known

## Section 15 - Regulatory Information

Safety, health and environmental regulations/legislation specific for the substance or mixture

National Regulations Australia

See section 8 for national exposure control parameters.

### Standard for the Uniform Scheduling of Medicines and Poisons

Classified as a scheduled poison according to the Standard for Uniform Scheduling of Medicines and Poisons.

| Component                                                                       | Standard for the Uniform Scheduling of Medicines and Poisons                                                                                                                                                                                        |
|---------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Silanamine,<br>1,1,1-trimethyl-N-(trimethylsilyl)-, lithium<br>salt - 4039-32-1 | Schedule 2 listed<br>Schedule 4 listed - for therapeutic use except: when included in Schedule 2, when present as an<br>excipient in preparations for dermal use containing <=0.25% of Lithium, or in preparations containing<br><=0.01% of Lithium |

### Australian - Illicit Drug Precursors/Reagents Substance List

This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list.

### Chemicals of Security Concern

This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern

National pollutant inventory Not applicable

### Prohibition or notification/licensing requirements

Shown below are details of specific prohibition/notifications or licencing requirements when they apply.

This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction.

### International Inventories

| Component                                                        | AICS | NZIoC | EINECS    | ELINCS | TSCA | DSL | NDSL | PICCS | ENCs | ISHL | IECSC | KECL |
|------------------------------------------------------------------|------|-------|-----------|--------|------|-----|------|-------|------|------|-------|------|
| Silanamine,<br>1,1,1-trimethyl-N-(trimethylsilyl)-, lithium salt | -    | X     | 223-725-6 | -      | X    | -   | X    | X     | X    | X    | -     | -    |

**Legend:** X - Listed. '-' - Not Listed. **KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

### International Regulations

**Ozone Depletion Potential** This product does not contain any known or suspected substance

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Rotterdam Convention (PIC)** Not applicable



**Basel convention on the control of transboundary movements of hazardous wastes and their disposal**  
Not applicable.

| Component                                                     | CAS No    | OECD HPV       | Restriction of Hazardous Substances (RoHS) | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|---------------------------------------------------------------|-----------|----------------|--------------------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Silanamine, 1,1,1-trimethyl-N-(trimethylsilyl)-, lithium salt | 4039-32-1 | Not applicable | Not applicable                             | Not applicable                                                                            | Not applicable                                                                           |

**Authorisation/Restrictions according to EU REACH**

Not applicable

## Section 16 - Other Information

### Legend

**AICS** - Australian Inventory of Chemical Substances  
**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory  
**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List  
**IECSC** - Chinese Inventory of Existing Chemical Substances  
**PICCS** - Philippines Inventory of Chemicals and Chemical Substances  
**TWA** - Time Weighted Average  
**IARC** - International Agency for Research on Cancer  
**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association  
**MARPOL** - International Convention for the Prevention of Pollution from Ships  
**NZS 5433:2012** - Transport of Dangerous Goods on Land  
**LD50** - Lethal Dose 50%  
**EC50** - Effective Concentration 50%  
**WEL** - Workplace Exposure Limit  
**DNEL** - Derived No Effect Level  
**POW** - Partition coefficient Octanol:Water  
**vPvB** - very Persistent, very Bioaccumulative  
**VOC** - (Volatile Organic Compound)

**NZIoC** - New Zealand Inventory of Chemicals  
**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances  
**ENCS** - Japanese Existing and New Chemical Substances  
**KECL** - Korean Existing and Evaluated Chemical Substances  
**CAS** - Chemical Abstracts Service  
**ACGIH** - American Conference of Governmental Industrial Hygienists  
**PNEC** - Predicted No Effect Concentration  
**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code  
**ADG** Australian Code for the Transport of Dangerous Goods by Road and Rail  
**OECD** - Organisation for Economic Co-operation and Development  
**LC50** - Lethal Concentration 50%  
**ATE** - Acute Toxicity Estimate  
**RPE** - Respiratory Protective Equipment  
**NOEC** - No Observed Effect Concentration  
**BCF** - Bioconcentration factor  
**PBT** - Persistent, Bioaccumulative, Toxic

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>  
 Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.  
 Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.  
 First aid for chemical exposure, including the use of eye wash and safety showers.

### Revision Date

17-Nov-2022

### Revision Summary

Not applicable.

**This Safety Data Sheet (SDS) is prepared in accordance to and complies with the requirements of Safe Work Australia - Work Health and Safety Regulations (WHS Regulations).**

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**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**